

| 操作系统作业6

系统：Ubuntu22.04.5

结果：

(红色为输入，蓝色为file1，绿色为file2，黄色为file3)

```
HelloWorld(base) root@REFINE-LV:~# vim pipe.c
(base) root@REFINE-LV:~# gcc pipe.c -o pipe
(base) root@REFINE-LV:~# ./pipe
Hello World!123456@lvlebin666
(base) root@REFINE-LV:~# cat file1.txt
HelloWorldlvlebin(base) root@REFINE-LV:~# cat file2.txt
12345666(base) root@REFINE-LV:~# cat file3.txt
!@
(base) root@REFINE-LV:~#
```

源码：

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <ctype.h>
#include <string.h>

int main() {
    int pipe_fd[2];
    if (pipe(pipe_fd) == -1) {
        perror("管道创建失败");
        exit(EXIT_FAILURE);
    }

    pid_t pid = fork();
    if (pid < 0) {
        perror("进程创建失败");
        exit(EXIT_FAILURE);
    }

    if (pid > 0) { // 父进程
        close(pipe_fd[0]);
        char buffer[BUFSIZ];
        while (fgets(buffer, sizeof(buffer), stdin) != NULL) {
```

```

        write(pipe_fd[1], buffer, strlen(buffer));
    }
    close(pipe_fd[1]);
    wait(NULL); // 等待子进程结束
} else { // 子进程
    close(pipe_fd[1]);
    FILE *f1 = fopen("file1.txt", "w");
    FILE *f2 = fopen("file2.txt", "w");
    FILE *f3 = fopen("file3.txt", "w");
    if (!f1 || !f2 || !f3) {
        perror("fopen");
        exit(EXIT_FAILURE);
    }
    char buffer[BUFSIZ];
    ssize_t bytes_read;
    while ((bytes_read = read(pipe_fd[0], buffer, sizeof(buffer))) > 0) {
        for (ssize_t i = 0; i < bytes_read; ++i) {
            char c = buffer[i];
            if (isalpha(c)) {
                fputc(c, f1);
            } else if (isdigit(c)) {
                fputc(c, f2);
            } else {
                fputc(c, f3);
            }
        }
    }
    fclose(f1);
    fclose(f2);
    fclose(f3);
    close(pipe_fd[0]);
    exit(EXIT_SUCCESS);
}

return 0;
}

```